

NB_EditPostR1: Notebook Changes Since Report 1 Submission

Baseline (as submitted with Report 1): TCX3901_Group5_DhitaYasaningrum_Report1_Notebook.html (folder TCX3901_Group5_DhitaYasaningrum_Report1, exported 12 Sep 2026), 121 cells.

Current: SellerReliability_dhy.ipynb, 146 cells since 8 Oct 2026, 7:33 pm (Tables 1 to 3 compare the 27 Sep save, 131 cells; Table 4 records the text edits of 8 Oct; Table 5 records new cells added from the ExtraNotes_ForRef tracker).

Checked: 8 Oct 2026, read-only (neither file was changed). Every cell was matched by content; cell numbers below count from 1 at the top, the same numbering used in ExtraNotes_ForRef.

Summary: all code from the submitted notebook is unchanged except one figure title (Figure 4 renamed 4a). Ten cells were added (+1 cell converted from markdown to code), and four markdown cells had punctuation-only edits. No number reported in Report 1 changed. The only output that changed is the Section 10 summary, because the new drop step (cell 105) now removes 131 sellers with no usable data.

Table 1: What changed

#	Section	Report 1 notebook (12 Sep)	Current notebook (27 Sep)	Type	Note
1	9. Cleaning	<i>Not in submitted notebook.</i> clean_seller_table kept 3,076 sellers, including 128 with no lead time and 3 with no review (NaN).	New code cell 105 "Decision: drop, not impute". Drops sellers with no usable lead-time or review data. Output: "Dropped 131 sellers ... clean_seller_table now has 2,945 sellers".	Added (code)	Matches Report 1 text (2,945 ratings sellers). ForRef To Do 8 = DONE.
2	10. Sanity checks	Cell 106 output: NaN counts 128 / 3; describe() on 2,948 lead-time rows and 3,073 review rows (mean review 4.04, nps_proxy 41.46).	Cell 107 (same code): 0 NaN everywhere; describe() on 2,945 rows (mean lead time 12.28 d, SD 7.20; mean review 4.14; nps_proxy 46.72).	Output changed	Knock-on effect of item 1, code unchanged. Note: the 12.3 / 7.2 in Q37 come from the 2,948 regression sellers (cell 74), not this table.
3	11. Save	Cell 108 was a markdown cell holding the save code (never ran; relative path "clean_seller_table.csv").	Cell 109 is a code cell, run. Saves to C:\Users\dhita\OneDrive\Desktop\TCX3901_db\clean_seller_table.csv. Output: 2,945 rows x 10 columns.	Converted	ForRef To Do 9 = DONE.
4	12.1 Regression baseline	<i>Not in submitted notebook</i> (Report 1 has only Baselines A and B).	New code cell 114: Baseline C, median lead time of the last 30 training days. Output: test mean 8.80 d; last-30-day median 9.06 d; MAE 2.81, RMSE 3.99.	Added (code)	Source of slide 9 Baseline C.
5	12.1	<i>Not present.</i>	New markdown cell 115 explaining Baseline C ("...MAE 2.81 / RMSE 3.99, the real bar for Report 2 models").	Added (text)	Wording now stronger than the presentation (see Table 3, d).
6	12.1	<i>Not present.</i>	New code cell 116: sensitivity check, windows 30 / 60 / 90 days x mean / median. MAE / RMSE: 30 mean 3.89 / 4.73; 30 median 2.81 / 3.99 ; 60 mean 4.12 / 4.93; 60 median 2.89 / 4.02; 90 mean 5.50 / 6.19; 90 median 3.56 / 4.46.	Added (code)	Backs Q&A Q27 (30 days not chosen as "best").

#	Section	Report 1 notebook (12 Sep)	Current notebook (27 Sep)	Type	Note
7	12.1	<i>Not present.</i>	New markdown cell 117: 30- and 60-day medians both give MAE about 2.8 to 2.9; Report 2 will choose the window on a validation period, not the test set.	Added (text)	
8	12.2 Classification baseline	<i>Not present</i> (accuracy and recall were printed in cell 114, which is unchanged).	New code cell 120: classification_report + confusion-matrix heatmap. Accuracy 0.8910; poor-class precision / recall / F1 = 0 / 0 / 0; support 17,583 not poor, 2,152 poor (19,735 test orders).	Added (code)	Source of slide 10 figures (89.1%, 2,152).
9	12.2	<i>Not present.</i>	New markdown cell 122: baseline is order level as a first pass; the real target needs a seller-month panel and will be re-measured before Report 2.	Added (text)	Matches slide 3 / slide 10 wording.
10	13. EDA	Cell 121: "Figure 4. Mean lead time by category (top 10 by item volume)".	Cell 128: renamed "Figure 4a" in the comment and chart title. Code otherwise identical.	Renamed	Slide 8 footnote cites Figure 4a.
11	13. EDA	<i>Not present.</i>	New code cell 129: Figure 4b, box plot of lead time by top-10 category (companion to 4a).	Added (code)	
12	14. Feature relationships (new section)	<i>Not present.</i>	New cells 130 to 131: heading "14. Feature relationships (Report 2 prep)" + scatter matrix of 5 seller features (avg_lead_time_days, on_time_rate, n_orders_lead, avg_review_score, nps_proxy).	Added (section)	Report 2 preparation; not one of the 4 Report 1 figures.
13	Text only	Cells 49, 104, 112, 115 used long dashes.	Cells 49, 104, 113, 121: dashes replaced with ", " ". " ":" "->" or "—>". Cell 104 also lost a minus sign: "(-8;" became "(8;".	Punctuation	No numbers changed. See Table 3 (a, b).

Table 2: What did not change

Area	Finding
Code	All code cells from the submitted notebook are identical in the current notebook, apart from the Figure 4 to 4a title (item 10).
Outputs	Same numbers throughout (for example 99,441 orders; 547 duplicate-review orders; 202 conflicting; 141 canceled; 6 canceled with a delivery date; 2,948 regression sellers; Baselines A and B; 89.1% accuracy). Some tables look different only because the HTML export shows them as formatted tables and the notebook stores them as text.
Headings	The "¶" marks after headings in the HTML are export anchors, not edits.
Cell mapping	Old 1 to 104 = new 1 to 104. Old 105 to 107 = new 106 to 108. Old 108 = new 109 (converted). Old 109 to 111 = new 110 to 112. Old 112 = new 113. Old 113 = new 118. Old 114 = new 119. Old 115 = new 121. Old 116 to 120 = new 123 to 127. Old 121 = new 128.
Tracker items	None of the ExtraNotes_ForRef notebook fixes (items 9 to 15, 17 to 19, 22, 23) is in the notebook yet; the notebook has not changed since 27 Sep.

Table 3: Small issues spotted on 8 Oct (ExtraNotes_ForRef item 24): all fixed, see Table 4

	Cell	Issue	Suggested fix
a	Cell 104 (Cleaning results table)	"96,470 with a usable lead time (8; ..." lost its minus sign.	Change "(8;" back to "(-8;" (96,478 to 96,470).
b	Cell 113 (regression baseline note)	"—>" appears three times ("The grand-mean baseline —> predicting ...").	Use ":" or "→", e.g. "The grand-mean baseline: predicting ...".
c	Cell 121 (classification note)	"->" used twice.	Optional: "→" or reword with a colon.
d	Cell 115 (Baseline C note)	"...the real bar for Report 2 models" is stronger than the presentation, which says C is not the final benchmark until A, B and C are compared on the same sellers.	Suggested: "...a strong simple benchmark for Report 2, to be confirmed once A, B and C are recomputed on the same eligible sellers."

Table 4: Edits made by Dhy on 8 Oct 2026 (saved 1:48 pm, 1:53 pm and 1:58 pm, checked in the saved file)

Markdown text and one code comment only: no code logic, outputs or numbers changed, so nothing needs re-running. Cell count was still 131 at that point (new cells were added later, see Table 5). ☑ = confirmed in the saved .ipynb.

Cell	Where	Before (27 Sep)	After (8 Oct)	Change	OK
104	9. Cleaning results table, row "Restrict to valid delivered orders"	96,470 with a usable lead time (8; the 6 canceled-with-a-date orders...	96,470 with a usable lead time (-8; the 6 canceled-with-a-date orders...	Minus sign restored (item 24a)	☑
104	9. Text under the table	...3 have no valid review record — every one of their orders...	...3 have no valid review record: every one of their orders...	Dash removed	☑
104	9. Text under the table	...not a bug — it needs a stated decision in Report 1...	...not a bug: it needs a stated decision in Report 1...	Dash removed	☑
104	9. Benchmark check	...below a 4.0 average review score — these two counts...	...below a 4.0 average review score: these two counts...	Dash removed	☑
113	12.1 Regression baseline note	The grand-mean baseline —> predicting the train-period average lead time (13.56 days) for every seller — scores...	The grand-mean baseline (predicting the train-period average lead time, 13.56 days, for every seller) scores...	Arrow removed (item 24b)	☑
113	12.1	A seller-aware baseline —> predicting each seller's own train-period average instead —> improves...	A seller-aware baseline (predicting each seller's own train-period average instead) improves...	Arrows removed (item 24b)	☑
113	12.1	...not order level — an earlier draft...	...not order level: an earlier draft...	Dash removed	☑
115	12.1 Baseline C note	...MAE 2.81 / RMSE 3.99, the real bar for Report 2 models.	...MAE 2.81 / RMSE 3.99, a strong simple benchmark for Report 2, to be confirmed once A, B and C are recomputed on the same eligible sellers.	Wording matches Presentation 1 (item 24d)	☑
121	12.2 Classification note	...which is "not poor" -> scores 89.1%... / ...imbalanced target -> the baseline...	...which is "not poor", scores 89.1%... / ...imbalanced target: the baseline...	Arrows removed (item 24c)	☑
122	12.2 Seller-month note	...poor-rating outcome in month t+1 — not a single row per seller.	...poor-rating outcome in month t+1, not a single row per seller.	Dash removed	☑
104	9. Text under the table (saved 1:53 pm)	Of those 3,076 sellers, 128 have no valid lead-time/on-time record and 3 have no valid review record: every one of their orders was either multi-seller-only, undelivered, or unreviewed. ... it needs a stated decision in Report 1 (drop those sellers, or impute with a category-level average) rather than being silently filled in.	19 sellers are not in the table: 17 only sold in multi-seller orders, and 2 had single-seller orders that were never delivered or reviewed. Of the 3,076 in the table, 128 have no delivered single-seller order (no valid lead time) and 3 have no reviewed one. Together, 147 sellers (19 + 128) have no valid lead time, leaving 2,948 for the regression: the same sellers as Section 7. ... Report 1 drops these sellers rather than imputing them (next cell).	Inaccurate sentence corrected; decision now stated (ForRef item 15c)	☑
113	12.1 (saved 1:53 pm)	...lead time, 13.56 days, for every seller... (double space)	...lead time, 13.56 days, for every seller...	Double space removed	☑

Cell	Where	Before (27 Sep)	After (8 Oct)	Change	OK
105	9. Drop decision, code comment only (saved 1:58 pm)	# Decision (Report 1 methodology): drop, not impute — these sellers have no underlying # orders/reviews to derive a real number from (every order was multi-seller-only, # undelivered, or unreviewed), so there's nothing genuine to fill in.	# Decision (Report 1 methodology): drop, not impute. These sellers have no underlying # delivered single-seller order (128) or no reviewed one (3), so there is # no genuine number to fill in.	Comment matches cell 104; not re-run, output kept ("Dropped 131 ... 2,945")	☒
106	10. Sanity checks intro (saved 1:58 pm)	No leakage columns went into clean_seller_table — it holds only seller_id...	No leakage columns went into clean_seller_table: it holds only seller_id...	Dash removed	☒

Nothing left open from this review. The "—" still in cell 104 is the empty Before cell of the last table row, not a sentence dash. Cell 105 was later re-run in order with the rest of the notebook (2:13 pm, see Table 5) and still prints "Dropped 131 ... 2,945".

Table 5: Tracker cells added to the notebook (ExtraNotes_ForRef items)

Each new cell moves every cell below it down by one; cell numbers in Tables 1 to 4 are as they were at the time, and the ExtraNotes_ForRef tracker keeps the current numbers. ☒ = confirmed in the saved .ipynb.

Item	Saved	Where	What it prints	Check	OK
19	8 Oct 2026, 2:13 pm	NEW code cell 128 when added, right after Figure 3, Section 13 (now cell 140, after items 11, 14, 23, 15a, 15b, 12, 10, 17, 18, 9 and 22 were added above it)	Grouped orders-per-seller counts behind Presentation 1 slide 7: 1 = 571, 2 = 353, 3 = 220, 4-5 = 299, 6-9 = 381, 10-19 = 453, 20-49 = 389, 50-99 = 219, 100-199 = 119, 200+ = 91. Total 3,095; fewer than 10 orders 1,824 (58.9%); median 6.	Kernel restarted (base, Python 3.13.9) and cells 1 to 128 run in order (run numbers 2 to 80). Re-run outputs unchanged, e.g. cell 105 "Dropped 131 ... 2,945", Baseline C MAE 2.81 / RMSE 3.99. Old Figure 4a onwards now cells 129 to 132 (not re-run, outputs kept).	☒
23	8 Oct 2026, 2:17 pm	NEW code cell 75, right after cell 74 (seller_lead_summary), Section 7.2	Sellers: 2,948 mean 12.3 d SD 7.2 d mean + 1 SD = 19.5 d. Above 20 days: 230 sellers (7.8%). Backs Q37 and the script (20 days is about mean + 1 SD).	Run on its own in the live session (run number 81). 230, not 229, because the 6 canceled orders are still in seller_order until item 13; after item 13 it prints 229 (also 7.8%). Every cell from the old 75 onwards moved down by one; ExtraNotes_ForRef tracker renumbered.	☒
15b	8 Oct 2026, 2:21 pm	NEW code cell 105 when added (now 106), right after the cell that builds clean_seller_table and before the Cleaning results table, Section 9	All sellers: 3,095 in clean_seller_table: 3,076 missing: 19 = 17 only in multi-seller orders + 2 never delivered or reviewed. In the table but no valid lead time: 128 no review: 3. No valid lead time in total: 147 -> regression sellers 2,948 (Section 7: 2,948).	Run after Execute Above Cells (run numbers 82 to 149), so clean_seller_table was rebuilt at 3,076 before the drop. Drop cell (now 107) not re-run; saved output still "Dropped 131 ... 2,945". Backs Q13 / Q14 and the cell 106 paragraph.	☒

Item	Saved	Where	What it prints	Check	OK
15a	8 Oct 2026, 2:24 pm	NEW markdown cell 78, end of Section 7 (after the 7.3 mean/median cell, before Section 8)	Note: 2,948 of the 3,095 sellers have a valid lead time (regression population; reviews not required). 147 without: 128 never delivered, 17 multi-seller only, 2 never delivered or reviewed. The 3 sellers with no review stay here and only drop out of the ratings table (2,945).	Text only, no run needed. Written without cell numbers so it stays correct if more cells are added. Placed at the end of Section 7 instead of directly after item 23; reads as a summary of the section.	☒
12	8 Oct 2026, 2:29 pm	NEW code cell 121, end of Section 12.1 (after the sensitivity-check note, before the 12.2 heading)	Cut-off: 2018-05-25 21:15:02 delivered after it: 1,998 of 76,160 (2.6%). Strict grand mean (orders delivered by the cut-off only): 13.33, MAE 5.34, RMSE 6.04 (vs Baseline A 5.53 / 6.21).	Run on its own in the live session (run number 150); numbers match the raw-data check. New variable names (tr, late, known, g) are not used by any later cell, so nothing else changes. Backs Q&A Q7 (leakage in the baseline); Report 2 will train only on orders delivered before the cut-off.	☒
10	8 Oct 2026, 2:32 pm	NEW code cell 124, right after the classification baseline (cell 123), Section 12.2	All reviewed: 98,673 orders, poor share 14.7%. Train: 78,938, 15.6%. Test: 19,735, 10.9% (2,152 poor ratings).	Run on its own in the live session (run number 151); numbers match the raw-data check. Only prints; no variables changed, so the classification report and confusion matrix below are unchanged. Added an All reviewed line beyond the tracker version. Backs slides 6 and 10 and Q&A Q39 (85.3% vs 89.1%).	☒
22	8 Oct 2026, 2:35 pm	NEW code cell 132, right after Figure 2 (cell 131), Section 13	Reviewed orders: 98,673 mean review score: 4.086 share of 4 or 5: 77.1%.	Run on its own in the live session (run number 152); matches the raw-data check. Only prints. Order-level background for the slide 6 backup line and Q&A Q47 / Q49, not the seller-month cut-off.	☒
11	8 Oct 2026, 2:44 pm	NEW code cell 60, right after the Step 1–3 join cell (cell 59), Section 4	Orders with items: 98,666. Orders with no items: 775 (unavailable 603, canceled 164, created 5, invoiced 2, shipped 1).	Run on its own in the live session (run number 153); matches the raw-data check. Only reads the data; orders_no_items is not used elsewhere, so the Step 4 cell and everything after are unchanged. Backs Q&A Q11 (98,666 vs 99,441). Every cell from the old 60 onwards moved down by one (item 13 is now in cell 73).	☒
14	8 Oct 2026, 2:55 pm	NEW code cell 68 + markdown cell 69, end of Section 5 (after the "Two things wrong with the raw data" note, before Section 6)	Sellers per order: 1 = 97,388 2 = 1,219 3 = 54 4 = 3 5 = 2. Multi-seller orders: 1,278 -> order-seller pairs: 2,622. Note (cell 69): 97,388 single-seller orders = the "97,388 kept" pairs in Section 9 (97,388 + 1,278 = 98,666 orders with items); 1,219×2 + 54×3 + 3×4 + 2×5 = 2,622 = the "2,622 set aside".	Run on its own in the live session (run number 154); matches the raw-data check. Only prints; spo and multi are not used elsewhere. Backs Q&A Q10. Not to be confused with 95,201: single-seller orders that also have a valid lead time. Every cell from the old 67 onwards moved down by two (item 13 is now in cell 75).	☒
17	8 Oct 2026, 3:04 pm	NEW code cell 128, right after item 10 (cell 127), Section 12.2	Reviewed orders: 98,673 no delivery date: 2,843 multi-seller: 1,264 no items: 756. Overlap: all 756 no-item orders and 3 multi-seller orders also have no delivery date. In any of the three groups: 4,104 single-seller with a delivery date: 94,569.	Run on its own in the live session (run number 155); matches the raw-data check. Only prints; no_date, multi_s and no_items are not used elsewhere. Backs Q&A Q9 (98,673 vs 95,201) and the Report 2 wording note (item 16: only 1,264 of the 1,278 multi-seller orders have a review). 94,569 is the starting point of item 18.	☒

Item	Saved	Where	What it prints	Check	OK
18	8 Oct 2026, 3:13 pm	NEW code cell 129, right after item 17 (cell 128), Section 12.2	Classification: 98,673 regression: 95,201. Reviewed & in regression: 94,569 reviewed, not in regression: 4,104 (no items 756 + multi-seller 1,264 + single-seller no delivery date 2,084) in regression, no review: 632. Check: 95,201 – 632 + 4,104 = 98,673.	Run on its own in the live session (run number 156); matches the raw-data check. Only prints; variables renamed gap_no_items / gap_multi so they do not reuse the item 14 and 17 names. Full answer to Q&A Q9. First place the notebook prints 95,201 as a number.	☒
9	8 Oct 2026, 3:20 pm	NEW code cell 130, right after item 18 (cell 129), Section 12.2	Crosstab review_score × poor_rating: 1 → 11,365 True; 2 → 3,128 True; 3 → 8,133, 4 → 19,044, 5 → 57,003 False. Counts: False 84,180 True 14,493 (= 98,673; 14.7% poor).	Run on its own in the live session (run number 157); matches the raw-data check. Display only, no new variables. Crosstab used instead of the tracker's head(10) so both classes show. Shows the target definition (score 1–2 = poor) for Q&A Q48.	☒
Fig 3b	8 Oct 2026, 6:30 pm	NEW code cell 141, right after the grouped counts (cell 140), Section 13	Prints: fewer than 10 orders 1,824 sellers (58.9%) 10+ orders 1,271 sellers (41.1%) total 3,095. Draws Figure 3b, the slide 7 chart (10 grouped bins, orange < 10 orders, grey 10+, median 6), in Times New Roman; saves fig3b_orders_per_seller_grouped.png.	Not a tracker item; added so the slide 7 chart and the 1,271 / 41% are produced by the notebook. All numbers calculated, none typed in. Font set only for this chart. Figure 4a, 4b and Section 14 moved down by one (now cells 142–145).	☒
Fig 4a values	8 Oct 2026, 7:33 pm	NEW code cell 143, right after Figure 4a (cell 142), Section 13	The 10 category means with item counts: utilidades_domesticas (housewares) 10.9 ... ferramentas_jardim (garden tools) 13.7; Range: 10.9 to 13.7 days (difference 2.8). 6 categories above and 4 below the 12.6-day mean.	Not a tracker item; added so the slide 8 numbers are printed, not just bar lengths. Before running it, the whole notebook was re-run in order with Execute Above Cells (run numbers 161 to 252; Figure 4b and Section 14 below it were not re-run): all saved outputs unchanged (e.g. cell 111 still "Dropped 131 ... 2,945"). Figure 4b and Section 14 moved down by one (now cells 144–146). Category names still Portuguese (ExtraNotes_ForRef item 25).	☒